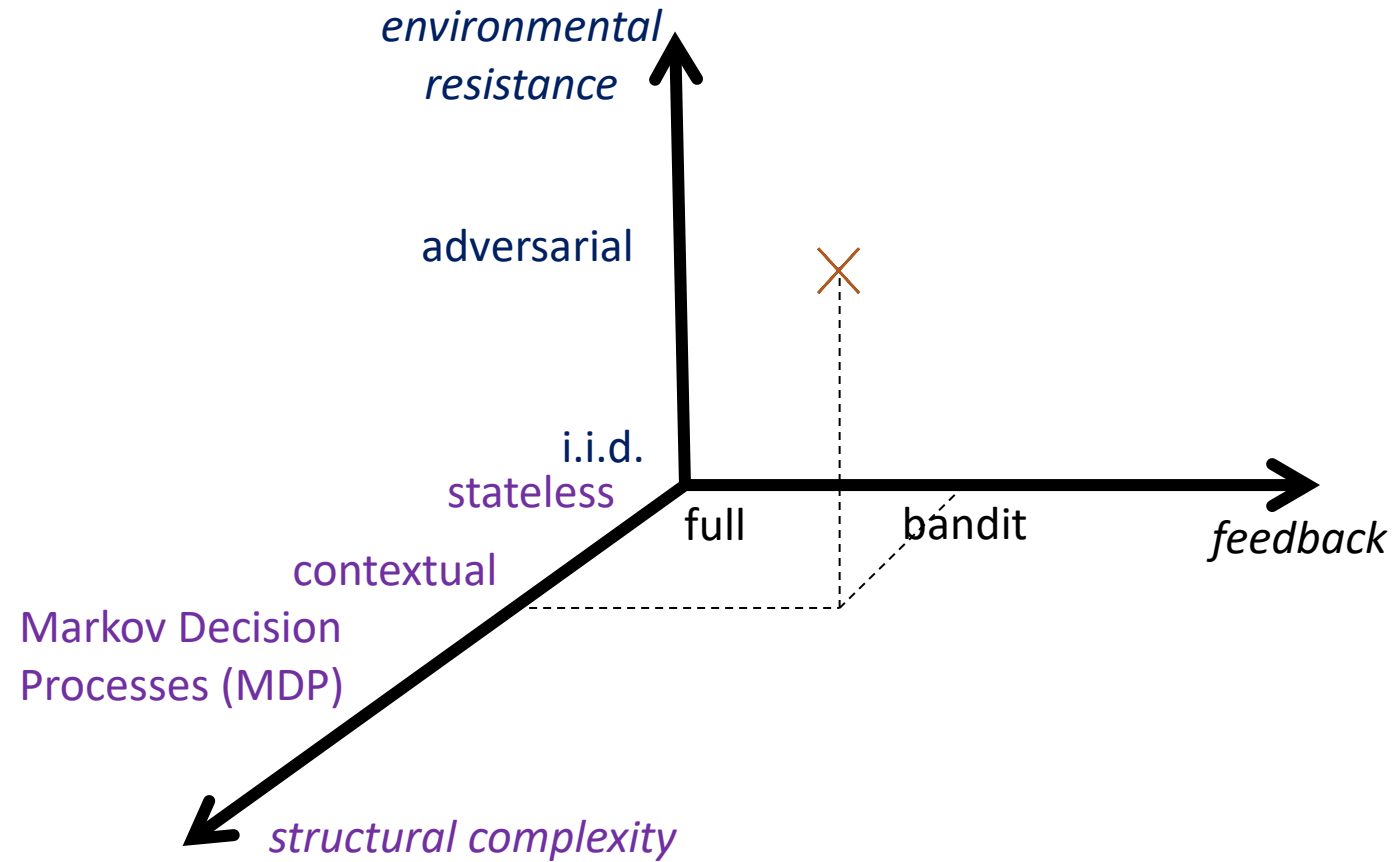


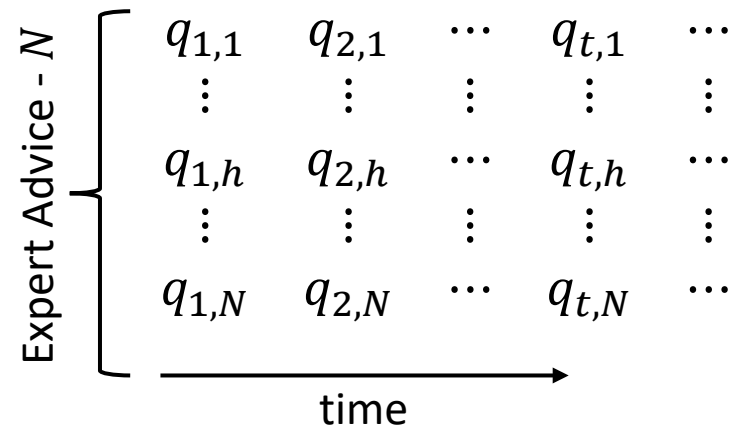
# Contextual Bandits

Yevgeny Seldin

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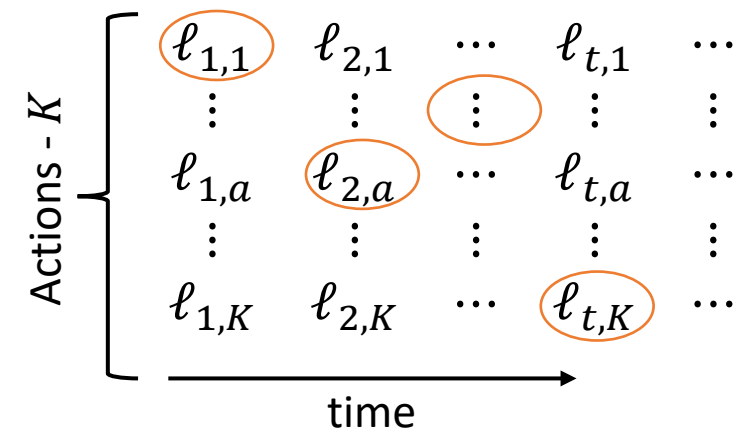
# Version #1: Bandits with Expert Advice



Game definition:

- For  $t = 1, 2, \dots$ 
  - Observe advice of  $N$  experts  $q_{t,1}, \dots, q_{t,N}$ 
    - where  $q_{t,h}$  is a distribution on actions  $\{1, \dots, K\}$
  - Play an action  $A_t$
  - Suffer and observe  $\ell_{t,A_t}$

Performance measure – regret:



$$R_T = \underbrace{\sum_{t=1}^T \ell_{t,A_t}}_{\text{Loss of the algorithm}} - \min_h \underbrace{\sum_{t=1}^T \sum_a q_{t,h}(a) \ell_{t,a}}_{\text{(Expected) loss of expert } h}$$

Deterministic  $q_{t,h}$  models a path through loss matrix

# Algorithm: EXP4

(Exponential Exploration Exploitation with Expert Advice)

- $\forall h: \tilde{L}_0(h) = 0$
- For  $t = 1, 2, \dots$ 
  - $\forall a: p_t(a) = \underbrace{\sum_h q_{t,h}(a)}_{\text{Advice}} \underbrace{\frac{e^{-\eta_t \tilde{L}_{t-1}(h)}}{\sum_{h'} e^{-\eta_t \tilde{L}_{t-1}(h')}}}_{\text{Weight of expert } h}$
  - $A_t \sim p_t$
  - [Observe  $\ell_{t,A_t}$ ]
  - $\forall a: \tilde{\ell}_{t,a} = \frac{\ell_{t,a} \mathbb{1}(A_t=a)}{p_t(a)}$
  - $\forall h: \tilde{\ell}_{t,h} = \sum_a q_{t,h}(a) \tilde{\ell}_{t,a}$
  - $\forall h: \tilde{L}_t(h) = \tilde{L}_{t-1}(h) + \tilde{\ell}_{t,h}$

- EXP4 Expected regret upper bound:

$$\mathbb{E}[R_T] \leq \sqrt{2 \underbrace{KT}_{\substack{\text{Price of} \\ \text{bandit} \\ \text{feedback}}} \underbrace{\ln N}_{\substack{\text{Complexity of} \\ \text{the comparator} \\ \text{class}}}}$$

- A different algorithm (based on regularization achieving  $O(\sqrt{KT})$  regret in stateless bandits) achieves

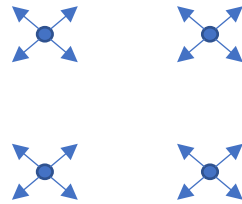
$$\mathbb{E}[R_T] = O\left(\sqrt{KT \ln_* \frac{N}{K}}\right)$$

- And there is a matching lower bound

# Version #2: Bandits with side information

Game definition:

- For  $t = 1, 2, \dots$ 
  - Observe side info (state)  $S_t$
  - Play an action  $A_t$
  - Suffer and observe  $\ell(A_t, S_t)$



Regret:

$$R_T = \underbrace{\sum_{t=1}^T \ell_t(A_t, S_t)}_{\text{Loss of the algorithm}} - \underbrace{\sum_{s \in \mathcal{S}} \min_a \sum_{t: S_t=s} \ell_t(a, s)}_{\substack{\text{Loss of the best action} \\ \text{in hindsight in state } s}} = \underbrace{\sum_{s \in \mathcal{S}} \min_a \sum_{t: S_t=s} \ell_t(a, s)}_{\text{Total loss assuming the best action in hindsight in each state}}$$

- Regret upper bound:

- Run the optimal  $O(\sqrt{KT})$  bandit algorithm in each state
- $T_s$  - the number of times state  $s$  appears in the sequence

$$\mathbb{E}[R_T] \leq \sum_{s \in \mathcal{S}} O(\sqrt{KT_s}) \leq O(|\mathcal{S}| \sqrt{KT|\mathcal{S}|}) = O(\sqrt{KT|\mathcal{S}|})$$

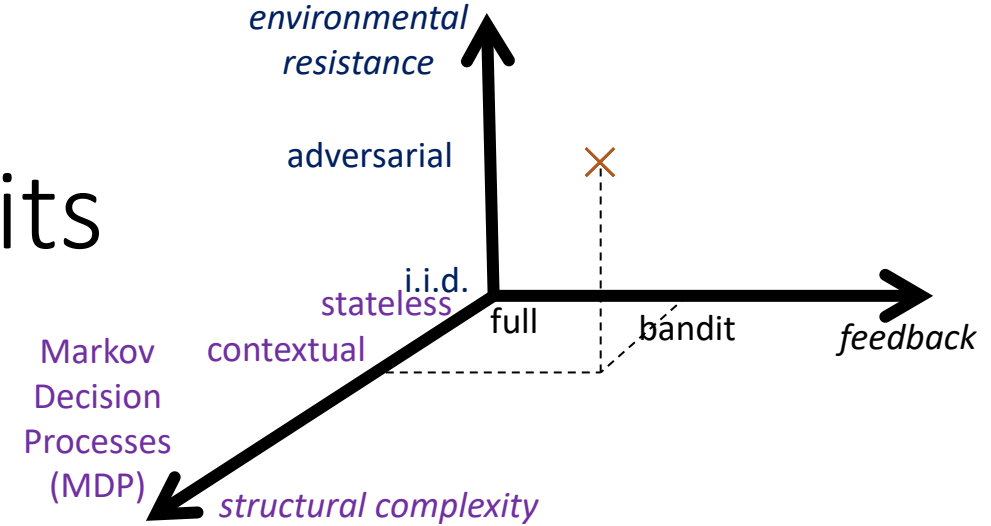
- Lower bound:

- Construct  $|\mathcal{S}|$  independent worst-case bandits played  $T/|\mathcal{S}|$  times each

$$\mathbb{E}[R_T] = \Omega \left( \underbrace{|\mathcal{S}|}_{\#(\text{bandits})} \underbrace{\sqrt{KT/|\mathcal{S}|}}_{\text{regret of each bandit}} \right) = \Omega(\sqrt{KT|\mathcal{S}|})$$

- $|\mathcal{S}|$  - structural complexity

# Summary – Contextual Bandits



- Bandits with Expert Advice

- EXP4 algorithm

- $p_t(a) = \sum_h \underbrace{q_{t,h}(a)}_{\text{Advice}} \underbrace{\frac{e^{-\eta_t \tilde{L}_{t-1}(h)}}{\sum_{h'} e^{-\eta_t \tilde{L}_{t-1}(h')}}}_{\text{Weight of expert } h}$

- $\mathbb{E}[R_T] \leq \sqrt{2KT \ln N}$

- Optimal algorithm:  $\mathbb{E}[R_T] = \theta \left( \sqrt{2KT \ln_* \frac{N}{K}} \right)$

- Bandits with side information

- $\mathbb{E}[R_T] = \theta \left( \sqrt{KT |\mathcal{S}|} \right)$

